Modeling Auckland Property Prices with Explainable AI

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***Abstract*—House prices affect lending, rates and major financial decisions, yet free online valuation tools give buyers a number without a reason. We build an explainable machine-learning model for Auckland property prices and test whether Auckland-specific external features (location, flood exposure, planning zone, aircraft noise, development activity and walking access) improve prediction beyond a property's own characteristics, and how the result compares with a traditional hedonic model and free online valuation estimates. We link 59,798 Auckland sales (January 2024 to August 2026) to map coordinates by probabilistic record linkage and train a fixed-configuration XGBoost model under five-fold, time-ordered cross-validation, so later sales never leak into earlier training folds. Adding 13 external features lowers test RMSE from $317,631 to $214,530 and MAPE from 18.2% to 11.3%, a statistically significant gain (bootstrap 95% CI excludes zero) that a two-stage residual model shows is mostly new information rather than something the property features already captured. Three attribution methods – permutation importance, SHAP and drop-one retraining – agree that almost all of this gain comes from the property's coordinates rather than the other 11 named features, a finding a theoretically correct joint-location explainer (GeoShapley) confirms. The model is far more accurate than a hedonic regression (MAPE 11.3% vs 20.9%); the comparison with free online valuation estimates is still in progress.**

***Index Terms*—house price prediction, explainable AI, SHAP, GeoShapley, gradient boosting, hedonic regression**

I. Introduction

For most New Zealand households, a home is the largest thing they will ever buy, and its value affects many decisions. Buyers and sellers use valuations to set prices, banks use them to decide how much to lend, and Auckland Council uses them to set rates. In Auckland, these valuations are hard to get right. Homes vary widely in size, age and condition, and the region itself varies too: some areas are close to the city centre or the coast, while others are exposed to flooding, aircraft noise or fast-changing development. A model that only looks at the house itself is therefore likely to miss a large part of its price.

Free online valuation tools give buyers a number but not a reason. A buyer cannot see whether an estimate is high because of the house, its location, or a flood risk the tool has picked up. This matters when the estimate is used to make a large financial decision.

Past research has looked at this problem from two directions. Hedonic studies of Auckland have measured the effect of single external factors on price, such as flood risk [1], school zones [2], leaky-building stigma [3] and social housing [4]. These models are easy to interpret, but each studies one factor at a time, and they are built to explain effects rather than to predict new sales [5]. Machine learning models predict prices more accurately than hedonic regression [6], [7], and SHAP [8] can show which features drive their predictions. However, we found no study that applies explainable machine learning to Auckland, and few that test how much a combined set of external features improves prediction for sales the model has not seen.

This paper addresses the following research questions:

- **RQ1:** Does adding Auckland-specific external features (location, flood, planning, noise, development and access) improve house price prediction compared with using only a property's own features?
- **RQ2:** How accurate and interpretable is our explainable model compared with a traditional hedonic model and free online valuation estimates for the same Auckland properties?

To answer these questions, we built a dataset of 59,798 single-home Auckland sales from January 2024 to August 2026. We linked each sale to a map coordinate using probabilistic record linkage, and used the coordinate to add 13 external features. We then trained the same XGBoost model twice, once with and once without the external features, using time-ordered cross-validation so that the model always learns from earlier sales and is tested on later ones. A paired bootstrap tests whether the difference is real (H1), and a two-stage model tests whether external features explain errors that property features cannot (H2). Three attribution methods show which external features matter. Finally, we compare the model with a hedonic regression on the same folds (H3) and with the council's capital value on a matched subset of sales.

Our main contributions are:

1) **A linked Auckland dataset.** We join council sales records to LINZ addresses using probabilistic record linkage, with a 96.7% match rate, and add flood, planning, noise, development and walking-distance features to each sale. We describe each cleaning step so that the dataset can be rebuilt.
2) **A fair test of external data.** By keeping the model and its settings fixed and training only on past sales, we isolate the effect of the external features. They reduce RMSE from $317,631 to $214,530 and MAPE from 18.2% to 11.3%, a significant improvement, and on their own they explain 26.6% of the error left by a property-only model.
3) **Evidence on what drives the gain.** Three attribution

methods agree that almost all of the benefit comes from the property's coordinates. The other external features add little once location is known, because they largely describe the same thing.

4) **Comparison with existing valuations.** The model is far more accurate than a hedonic regression (MAPE 11.3% vs 20.9%), but still slightly less accurate than the council's capital value on matched sales (11.0% vs 10.5%), without using any valuer input.

The rest of this paper is organised as follows. Section II reviews related work. Section III describes the data and how it was cleaned and linked. Section IV explains the experimental design. Section V presents and discusses the results. Section VI sets out the limitations, and Section VII concludes and suggests future work.

II. Related Work

A. Hedonic models and external factors in Auckland

Hedonic pricing treats a house as a bundle of characteristics, each with its own implicit price, which can be estimated by regressing sale price on those characteristics [9]. Hedonic models are mainly built to *explain* the effect of each factor, not to predict new sales as accurately as possible [5]. Auckland studies using this approach show that factors outside the property matter. Adding valuer-defined submarkets to a hedonic model improved prediction more than replacing it with more advanced spatial models [10]. Flood-prone homes on the North Shore sold for about 6.2% less before flood maps were published and 2.3% less afterwards [1]. Being in the zone of a popular school raised prices [2], while leaky-building stigma lowered house prices by about 5% [3], and nearby social housing lowered prices by 1.7–23% [4]. Studies outside New Zealand also show that ignoring the link between nearby sales can overstate the effect of an external factor such as flooding [11].

These studies have two limits. First, each one looks at a single external factor, so they do not show how much a combined set of external data improves prediction. Second, the models are linear, and the form of each effect (for example, how price changes with distance) must be chosen by hand. Our work instead adds 13 external features to one model at the same time and measures their combined effect on prediction error for sales the model has not seen.

B. Machine learning versus hedonic models

Tree-based machine learning models learn non-linear effects and interactions directly from the data. Across many studies they predict prices more accurately than hedonic regression. A random forest reached a test MAPE of about 7% against 17% for regression on Ljubljana apartments [6], and an R^2 of 0.94 against 0.65 on Busan condominiums [7]. In Russia, a random forest also beat regression (MAPE 14.9% vs 18.3%), although its error was much higher for larger apartments [12], so a single overall error can hide weak spots. Gradient-boosted trees such as XGBoost [13] perform similarly well. Soltani *et al.* [14] improved tree models further by adding a feature built from nearby recent sales, and stressed that such features must use only sales that happened *before* the sale being predicted.

Most of these comparisons change the model, the features, or both at once, so it is hard to tell how much of the gain comes from better data and how much from a better algorithm. We keep the algorithm and its settings fixed and change only the feature set, so any difference comes from the external data. We also train only on past sales and test on later ones, which avoids leaking future information into the model.

C. External data and explainable models

Machine learning models are harder to interpret than hedonic models, which matters when buyers and lenders need to know why a home is valued as it is. SHAP [8] addresses this by splitting each prediction into the contribution of each feature. In Lisbon, adding open data on the surroundings reduced MAE by 8.24%, and SHAP showed that longitude and latitude were among the most important features [15]. In Sydney, SHAP showed that access to amenities affected price in a non-linear way that changed with distance and type of amenity [16]. In Beijing, location was the most important factor, and the effects of other features varied between areas [17]. SHAP, however, describes how the model uses each feature, not what causes prices to change, and it cannot show on its own that a model predicts well.

These studies use SHAP to rank features within one model. When features carry overlapping information, as coordinates and location-based features do, a single ranking can be misleading. We therefore first test whether external features improve prediction, and only then explain the model. To see how much each external feature matters, we use three methods (permutation importance, SHAP and drop-one retraining) that react differently to overlapping features, and we confirm with a separate residual test that the external features add information the property features do not.

D. Summary

To our knowledge, no published study has applied explainable machine learning to Auckland house prices, and few have tested the joint predictive value of external features in a single model. Our work addresses both gaps. It also compares the model with a hedonic model and with Auckland Council's capital value, the valuation most Auckland homeowners already see.

III. Data

A. Data sources

Our main data is the Auckland Council sales file [18]. It lists 92,010 property sales from January 2024 to August 2026, with 41 columns that describe the sale (price, date, sale type) and the property (land area, floor area, building age, condition, view, and so on).

We added external data from five sources: flood plain, flood-sensitive area and Unitary Plan zone maps from Auckland Council [19]; the aircraft noise overlay from the Unitary Plan; new dwellings consented in each statistical area

TABLE I
FILTERS APPLIED TO THE SALES DATA, IN ORDER

Filter	Sales left
All parsed sales	92,010
Residential, rural or lifestyle zoning	80,070
One unit of use	77,370
Freehold	76,395
One property per sale	74,112
Arm's-length market sale	61,244
Coordinate found	59,798

(SA2) from Stats NZ [20]; supermarket and road data from OpenStreetMap [21]; and bus stop locations from Auckland Transport [22].

B. Linking sales to coordinates

[Splink record linkage: purpose, LINZ address data, Fellegi–Sunter model and Splink, compared fields, match threshold, match rate, clerical review results.]

$$[matchweightequation] \tag{1}$$

Once each property had a coordinate, we placed it on the GIS maps to find whether it lies in a flood plain, flood-sensitive area or noise overlay, its distance to the nearest flood plain and flood-sensitive area, and its Unitary Plan zone. Walking distances to the nearest supermarket and bus stop were measured along the road network, not in a straight line.

C. Cleaning

We kept only sales that represent a normal market price for one home. Table I shows the filters, applied in order.

The zoning and unit filters remove commercial properties and buildings with several homes. The tenure filter removes leasehold sales, which sell for less because the buyer does not own the land. The one-property filter matters most: when several properties are sold together, the sale records one combined price but the features of only one property, so the model cannot learn from these rows. The arm's-length filter removes sales between related parties, which may not reflect market value. The last filter removes sales with no coordinate.

We also removed five columns that would leak the answer to the model. The net, chattels and other price columns add up to the sale price. The council's capital value and land value are themselves price estimates, so a model that uses them would copy the council's valuation instead of learning from property and location data.

D. Features

We built two feature sets (Table II). The *baseline* set has 22 features that describe only the property and the sale date; it contains no information about location. The *augmented* set adds 13 external features, for 35 in total. We also built driving distances, but dropped them because they were highly correlated with the walking distances.

Land area, the two floor areas and the two walking distances are right-skewed, so we applied $\log(1+x)$ to them. Categories are converted to integer codes, and missing numbers are filled with the median. Both steps are learned from the training data only (Section IV-B).

TABLE II
FEATURE SETS

Group	Features
Baseline (22)	
Size	land area, building floor area, total floor area, site coverage
Building	age, condition, construction type, deck, workshop or laundry, other improvements, garages, off-street parking
Site	contour, view, scope of view, zoning, property type, property use
Time	sale year, month, quarter
External (13)	
Location	latitude, longitude
Flood	in flood plain, in flood-sensitive area, any flood risk, distance to flood plain, distance to flood-sensitive area
Planning	Unitary Plan zone
Noise	in noise overlay, noise category
Development	new dwellings consented in the SA2 (past 12 months)
Access	walking distance to supermarket, walking distance to bus stop

IV. METHODOLOGY

All code is available online.[1]

A. Model

We use XGBoost [13], a gradient-boosted tree model that predicts the sale price in dollars. We chose it because it handles non-linear effects and mixed feature types well, and tree models have outperformed linear regression in earlier house price studies [6], [7].

All models use the same fixed settings: 500 trees, maximum depth 6, learning rate 0.05, 80% of rows and 80% of features sampled per tree, and random seed 42. We did not tune these settings for each model. If each model were tuned separately, part of any difference between them could come from tuning rather than from the features. Keeping the settings fixed means that differences between models come from the data only. Section IV-I checks how much accuracy this costs.

B. Training and testing

House prices change over time, so a model should only learn from sales that happened before the sales it predicts. Randomly shuffling the data would let the model learn from future sales, which makes results look better than they would be in practice [14].

We therefore sort the 59,798 sales by date and use five-fold expanding-window cross-validation (Fig. 1). The data is cut into six equal time blocks of about 9,966 sales. In fold k, the model trains on blocks 1 to k and is tested on block

[1] [https://github.com/ORG/REPO]. The main analysis is in `final.ipynb`; see the README for how to run it.

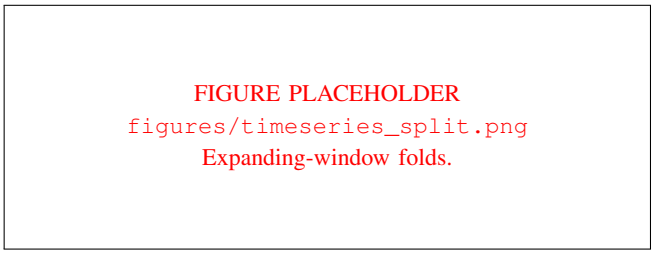


Fig. 1. Five-fold expanding-window cross-validation. Each fold trains on all earlier sales and tests on the next period.

$k+1$. Each fold therefore trains on more history than the one before. Every sale after the first block is tested exactly once, by a model that has never seen it. We do not keep a separate validation set because no settings are tuned.

Within each fold, all preprocessing is learned from the training blocks only: the category codes, the median values used to fill gaps, and the price limits described next. The test block is then processed with these training values.

Before training, we remove sales priced below the 1st percentile or above the 99th percentile of the training prices. The same limits are applied to the test block. This stops a few extreme sales from dominating the error. It also means our results describe the middle 98% of the market, not the very cheapest or most expensive homes. After this step, 48,848 test predictions remain across the five folds.

C. Evaluation metrics

For n test sales with actual price y_i and predicted price $\hat{y}_i$, we report three error measures:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \tag{2}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \tag{3}$$

$$\mathrm{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\frac{|y_i - \hat{y}_i|}{y_i}. \tag{4}$$

MAE is the typical error in dollars. RMSE gives more weight to large errors, which matter most for expensive homes. MAPE is the error as a percentage of price, so a \$100K error on a \$600K home counts more than the same error on a \$3M home. We combine the test predictions from all five folds and compute each measure once on the combined set.

D. Testing whether the improvement is real (H1)

To check that the augmented model is better and not just lucky, we use a paired bootstrap. We draw n test sales at random with replacement and compute the difference

$$\Delta^{(b)} = M^{(b)}_{\text{base}} - M^{(b)}_{\text{aug}} \tag{5}$$

for each metric M. Both models are scored on the same resampled sales, so the comparison is fair. We repeat this $B = 2{,}000$ times. The 2.5th and 97.5th percentiles of $\Delta^{(b)}$ give a 95% confidence interval. If the interval does not include zero, the improvement is statistically significant.

E. Two-stage residual test (H2)

The test above shows whether external features help, but the augmented model could still use them to learn something the property features already contain. The two-stage test checks whether external features add *new* information.

In stage 1, we take the baseline model's test predictions from Section IV-B and compute each sale's error $r_i = y_i - \hat{y}_i^{\text{base}}$. In stage 2, a new XGBoost model uses only the 13 external features to predict r_i. Stage 2 has no access to property features, so it can only succeed if location and external data explain what the property features missed.

Stage 2 uses its own five-fold expanding-window split over the 48,848 sales that have a stage-1 error, again sorted by date. The first block (8,141 sales) is used only for training, leaving 40,705 sales with a stage-2 prediction. We report the out-of-fold R^2 of stage 2 in Section V. A value above zero supports H2.

F. Which external features matter

For this analysis we use a single time split: the first 80% of sales for training (up to December 2025) and the last 20% for testing (December 2025 to August 2026), with the same price limits as before. We use three methods, because each one can be misleading on its own when features are correlated:

- **Grouped permutation importance.** We randomly shuffle all 13 external features together in the test set and measure how much the RMSE increases. We repeat this 20 times and report the average.
- **SHAP.** We compute SHAP values [8] on 500 test sales. Each SHAP value is the amount, in dollars, that a feature moves one prediction up or down. We report each external feature's average absolute SHAP value, and the external features' share of the total.
- **Drop-one retraining.** For each external feature, we retrain the model without it and measure the change in test RMSE.

These methods react differently when two features carry similar information. Permutation importance breaks all of them at once, so the model cannot fall back on a similar feature, which can overstate their importance. Drop-one retraining lets the model fall back on a similar feature, which can understate it. SHAP splits the credit between similar features. A conclusion that all three methods agree on is therefore more reliable than any single result.

G. Hedonic baseline

To compare with the traditional approach, we fit a hedonic model: a linear regression on the logarithm of price,

$$\ln y_i = \beta_0 + \mathbf{x}_i^{\top}\boldsymbol{\beta} + \varepsilon_i. \tag{6}$$

Following common hedonic practice, it uses a short, hand-picked list of features rather than all 35: land area, building floor area, site coverage, off-street parking, garage under main

roof, sale year, building age, condition, property type, zoning group, latitude, longitude, and flood risk. Categories are one-hot encoded. In early folds a category can be missing from the training data; any column that is constant in a fold's training data is dropped for that fold.

The model predicts log price, and turning this back into dollars with $\exp(\cdot)$ underestimates the average price. We correct this with Duan's smearing factor [23]:

$$\hat{y}_i = \exp\left(\hat{\beta}_0 + \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}\right) \cdot \frac{1}{m} \sum_{k=1}^{m} e^{\hat{\varepsilon}_k}, \tag{7}$$

where $\hat{\varepsilon}_k$ are the m training residuals. The hedonic model uses the same folds, price limits and metrics as XGBoost, so all results are directly comparable.

H. Online valuation benchmark

To test the second half of RQ2, we draw a sample of held-out, test-period sales and look up a free public valuation estimate for each one (e.g. homes.co.nz, propertyvalue.co.nz or trademe.co.nz), matched to the council record by address and confirmed against the actual sale date and price so we are comparing the estimate that was available *before* the sale, not one updated afterwards using the sale itself. We then compute the same RMSE/MAE/MAPE as the other models on this matched subset. [Sample size, match-confirmation results and the online-valuation error metrics.]

I. Hyperparameter check

To check whether the fixed settings were a reasonable choice, we tuned the augmented model once on the 80/20 split from Section IV-F. We tried 25 random combinations of tree count, depth, learning rate, sampling rates and regularisation, scoring each with a three-fold expanding-window split inside the training data. The test data was not used until the best combination had been chosen. We then compared the tuned and fixed models on the test data (Section V).

V. Experiments and Results

A. Does external data help? (H1)

Table III gives the pooled out-of-fold error over all five folds. Adding the 13 external features cuts RMSE by \$103,102, from \$317,631 to \$214,530 (a 32.5% relative reduction), MAE by \$77,075, and MAPE by 6.9 percentage points, from 18.2% to 11.3%. The paired bootstrap (Eq. 5, $B = 2{,}000$) gives a 95% confidence interval of \$99,920–\$106,136 for the RMSE improvement, which excludes zero, so the gain is real and not a result of how the test sales happen to fall. In practical terms, a MAPE of 11.3% on a typical \$1M Auckland home is an average error of roughly \$113,000, close to half the \$208,000 average error the baseline model makes; external data is doing real, not marginal, work.

TABLE III
Baseline (property only) vs. augmented (with external features), pooled out-of-fold over 5 folds, $n = 48{,}848$

Model	RMSE	MAE	MAPE
Baseline	\$317,631	\$208,319	18.2%
Augmented	\$214,530	\$131,243	11.3%

FIGURE PLACEHOLDER
figures/baseline_vs_augmented_metrics.png
Baseline vs. augmented RMSE/MAE/MAPE bar chart.

B. Do external features add new information? (H2)

A model could look like it needs external data only because the property features are weak, not because location genuinely carries new information. The two-stage test rules this out: a model using *only* the 13 external features explains 26.6% of the variance in what the property-only model gets wrong ($R^2 = 0.266$ on 40,705 held-out residuals), cutting the residual RMSE from \$314,149 to \$268,861. Since this second-stage model never sees a single property feature, this 26.6% must be information the property features do not already contain, supporting H2.

C. Which external features matter?

Table IV compares the three attribution methods from Section IV-F on the full augmented model (single 80/20 split, $n = 11{,}741$ test sales, RMSE \$190,061). Grouped permutation importance shows shuffling all 13 external features together increases RMSE by 148.7%. SHAP attributes 40.1% of total mean absolute SHAP to the external features, of which latitude (48.4%) and longitude (34.0%) alone account for 82.4%. Drop-one retraining agrees: removing latitude or longitude increases RMSE by \$36,700 and \$31,106 respectively, while removing any other single external feature changes RMSE by at most about \$1,100 (and for several, retraining without them is marginally *better*, within noise). All three methods – which react differently to correlated features (Section II) – agree on the same conclusion: almost all of the benefit of external data comes from the property's coordinates, and the other 11 named features add comparatively little once location is known, because they largely describe the same underlying thing (how desirable the area is).

TABLE IV
ATTRIBUTION OF THE TOP EXTERNAL FEATURES (SINGLE 80/20 SPLIT). DROP-ONE IS THE RMSE CHANGE FROM REMOVING THE FEATURE AND RETRAINING; NEGATIVE VALUES MEAN THE MODEL DID MARGINALLY BETTER WITHOUT IT.

Feature	SHAP share	Drop-one ΔRMSE
Latitude	19.4%	+$36,700
Longitude	13.7%	+$31,106
Dist. to flood-sensitive area	2.1%	−$98
Walking distance, supermarket	1.4%	−$1,022
Dist. to flood plain	1.0%	−$471
SA2 new dwellings consented (12m)	1.0%	−$1,131

FIGURE PLACEHOLDER
figures/shap_attribution_full_model.png
SHAP summary plot for the full augmented model.

Because splitting one geographic concept into two separate SHAP columns (latitude, longitude) is known to distort Shapley attribution for coordinate pairs [24], we repeat the SHAP analysis with GeoShapley, which treats location as one joint "GEO" player and also recovers location–feature interaction effects that ordinary SHAP cannot express. On the same 500-sale sample, the joint GEO effect is 18.8% of total mean absolute GeoShapley value, and GEO's interactions with other features (e.g. a feature's effect on price changing depending on where the property is) add a further 25.6%, for 44.5% combined. This is not directly comparable in scale to the 82.4% figure above (that denominator is external features only, not all 36 features), but it confirms the same conclusion with a theoretically valid number: location, taken as a single concept, is the dominant source of explanatory power among the external features, and part of what looked like "small, independent" effects from the other features in Table IV is really location changing how those features behave across Auckland.

D. Comparison with a hedonic model (RQ2)

Table V compares the hedonic regression (Section IV-G) with both ML models, pooled over the same 5 folds. The hedonic model is far less accurate than either ML model (MAPE 20.9% vs. 18.2% baseline and 11.3% augmented): a straight-line model, even with latitude and longitude included, cannot capture the non-linear and interacting effects a tree model learns directly from the data. In exchange, each hedonic coefficient has a direct, one-sentence interpretation (e.g. "+1 m^2 of floor area adds $X"), which is harder to get from SHAP, where a feature's contribution varies prediction by prediction. This is the accuracy/interpretability trade-off RQ2 asks about: our model keeps most of the hedonic model's explainability (via SHAP, drop-one and GeoShapley) while cutting its error by close to half.

TABLE V
HEDONIC REGRESSION VS. BOTH ML MODELS, POOLED OUT-OF-FOLD

Model	RMSE	MAE	MAPE
Hedonic (linear)	$683,994	$243,001	20.9%
ML, baseline	$317,631	$208,319	18.2%
ML, augmented	$214,530	$131,243	11.3%

TABLE VI
FIXED VS. TUNED HYPERPARAMETERS, SINGLE 80/20 SPLIT

Configuration	RMSE	MAE	MAPE
Fixed (used throughout)	$190,061	$121,158	10.68%
Tuned (25-iteration search)	$185,867	$117,814	10.36%

FIGURE PLACEHOLDER
figures/hedonic_three_way.png
Hedonic vs. baseline vs. augmented ML, pooled RMSE/MAE/MAPE.

E. Comparison with online valuation estimates (RQ2)

[Online valuation benchmark results: see Section IV-H – sample size, matched properties, and RMSE/MAE/MAPE on the matched subset, alongside the augmented model's error on the same sales.]

F. Does tuning help?

Table VI compares the project's fixed XGBoost configuration with the best of 25 randomly sampled configurations (Section IV-I). Tuning gives a small, real improvement: RMSE falls from $190,061 to $185,867 ($4,194, 2.2% relative), with similar gains in MAE and MAPE. The best configuration favours slightly deeper trees (depth 7 vs. 6), more of them (579 vs. 500), added regularisation, and a lower row-sampling fraction (0.67 vs. 0.8). A 2.2% gain is real but modest next to the 32.5% gain from adding external data (Section V-A) – over ten times larger. This supports keeping the hyperparameters fixed throughout the rest of this paper: it did not meaningfully understate what a well-tuned model could do, so the comparisons in Sections V-A–V-D are not an artefact of under-tuning one side.

VI. LIMITATIONS

Our price filter keeps only the middle 98% of each fold's training prices (Section IV-B), so our results describe the typical Auckland home and may not transfer to the cheapest or most expensive 1% of sales, where a council-assigned valuation or a hedonic model might behave very differently. Record linkage could not place a coordinate for every sale: 61,244 arm's-length single-property sales were available, and

59,798 (97.6%) were matched to a coordinate (Section III-B); if the unmatched 2.4% are systematically different (e.g. new subdivisions or non-standard addresses not yet in LINZ), our results could be slightly optimistic or pessimistic for that slice of the market, in a direction we have not separately quantified.

Our attribution methods measure association with the model's predictions, not causation: a high SHAP value for distance to a flood plain says the model leans on that feature, not that flood exposure *causes* a specific dollar change in price. GeoShapley's `check_additivity` test did not pass exactly on our model (the GEO/feature/interaction components do not sum to the prediction to floating-point precision); we believe this is a numerical-tolerance artefact of the package's fixed tolerance at this model's scale (500 trees, six-figure dollar predictions) rather than a logic error, but we did not confirm this beyond a synthetic sanity check, so the GeoShapley percentages in Section V-C should be read as approximate. Finally, we deliberately fixed the model's hyperparameters for a fair comparison across feature sets (Section V-F shows this cost at most 2.2% RMSE); a project optimising purely for accuracy could tune per feature set and would likely do marginally better than every number reported here.

VII. Conclusion and Future Work

This paper asked whether Auckland-specific external data improves house price prediction (RQ1), and how an explainable model built on it compares with a traditional hedonic model and free online valuations (RQ2). For RQ1, the answer is yes, and by a wide and statistically significant margin: adding 13 external features cuts RMSE by 32.5% and MAPE by 6.9 percentage points, and a two-stage test confirms this is new information the property features did not already contain. For the first half of RQ2, our model is far more accurate than a hedonic regression (MAPE 11.3% vs. 20.9%) while remaining explainable through SHAP, drop-one retraining and GeoShapley; the second half, comparison against free online valuation estimates, is still being completed (Section IV-H).

The clearest practical finding is that almost all of the value of "external data" in this project is really the value of *accurate location*: three independent attribution methods and a theoretically correct joint-location explainer all agree that latitude and longitude dominate, and that the other 11 named features (zoning, flood, noise, development, walking access) add comparatively little once location is known. For a team building a similar model with a limited data budget, this means investing in accurate geocoding is worth more than adding another hand-engineered location-adjacent feature.

Future work should (1) finish and report the online valuation benchmark; (2) use GeoShapley's location–feature interaction terms to ask whether features such as flood exposure matter more in specific parts of Auckland than in the city-wide average reported here; (3) investigate whether a small, carefully chosen set of named location proxies (e.g. distance to the CBD, a deprivation index) can substitute for raw coordinates when exact coordinates are unavailable – an exploratory side-experiment during this project found this depends heavily on *which* proxies are chosen, and deserves a dedicated study rather than the brief check we gave it; and (4) extend the record-linkage pipeline to recover the 2.4% of sales that currently have no matched coordinate, to check whether they change any of these conclusions.

Contributions

[Member 1 to contribution.]
[Member 2 to contribution.]
[Member 3 to contribution.]
[Member 4 to contribution.]
[Member 5 to contribution.]

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